

Benedict Julian Danandjaja Antoni

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PROFILE

Engineering Product Development undergraduate with experience in research, programming, CAD and fabrication. Interested in computational engineering, biomedical product development and materials science.

EDUCATION

RWTH Aachen University - Accepted Incoming Exchange Student, Winter Semester 2026/27 - Exchange under Faculty Agreement (Computer Science)	Aachen, Germany Oct 2026 – Mar 2027
Singapore University of Technology and Design - Bachelor of Engineering (Engineering Product Development) - SUTD Global Merit Scholarship	Singapore Sep 2024 – Present
Shanghai Jiao Tong University (School of Design) - Global Exploration Opportunities (GEO) “Healthy Ageing” International Summer School Workshop – The Sustainable Transformation of Food Design - Conceptualized 3D food printing solutions using CAD, exploring multi-nozzle designs for faster printing and app-based education to promote everyday adoption in sustainable home settings.	Shanghai, China Aug 2025 – Sep 2025
Nanyang Polytechnic - Diploma in Robotics & Mechatronics with Merit - Director’s List (Top 15% of Cohort), 2019 – 2022 - NYP Scholarship (Mid-term), 2020 & 2021 - Festo Gold Medal (2022) – Awarded to top graduate in the Diploma	Singapore Apr 2019 – May 2022
Montfort Secondary School GCE ‘O’ Levels Certificate	Singapore Jan 2015 – Dec 2018

WORK EXPERIENCE

Singapore Armed Forces National Service – Admin Support Assistant	Singapore Sep 2022 – Jul 2024
Certis Cisco Security Pte Ltd – Intern, Technical Assistant - Evaluated three open-source software tools for potential integration into the system. - Collaborated with the Artificial Intelligence department to support machine-learning data preparation. - Cleaned and compiled 600 data points weekly for 6 months. - As Digital Office intern: scheduled daily missions, wrote weekly performance reports, and recorded robot detection reports. - The robot’s three-month average mission effectiveness increased from 68.3% to 75.9%.	Singapore Sep 2021 – Feb 2022
Yamaha Pt. Sentosa Jaya Motor – Part-Time, Servicing Staff - Performed basic motorcycle servicing, including diagnostic checks and engine-oil changes. - Organised warehouse and replenished spare parts. - Ensured that safety measures were maintained during sales & events.	Bantaeng, Indonesia Sep 2019 – Oct 2019

ACADEMIC PROJECTS

Singapore University of Technology and Design, Undergraduate Research Opportunities Programme (UROP) Research Assistant	Sep 2025 – Apr 2026
1. Opportunities And Difficulties of Integrated Photonic Brain-inspired Computing - Supported the report background writing through journal research for: - Hardware-software co-optimization - Neural network models	
2. Inverse Design of Plasmonic Nanostructure Devices Via Topological Optimization - Assessed different algorithms and proposed a process for maximising waveguide reflection. - Validated the most impactful factors affecting the sample’s fitness function.	
3. Silicone Vacuum Bags for Composite Fabrication (PhD Project) - Assisted in optimising the fabrication process for faster and easier production of three prototypes. - Conducted 17 tests to determine the optimal speed, power and number of passes for the plasma surface-treatment machine. - Conducted and compiled results from up to 200 tests to evaluate the effectiveness of the surface treatment.	

CO-CURRICULAR ACTIVITIES

Capstone Community Project – The Mental Wellness Virtual Campaign

Assistant Student Organiser

Oct 2020 – Jan 2021

- Organised an art-based virtual campaign for 40 participants to raise awareness of student mental health and coping strategies during home-based learning.

National Police Cadet Corps (NPCC)

Feb 2015 – Apr 2018

- As Head of Administration, reported overall attendance to teaching staff, compiled necessary letters and forms, and assisted in planning weekly training schedule.
- Organised & led 2 training camps for junior members and 2 inter-school game events involving groups from 10 different schools.
 - Awarded Best Unit Cadet & Singapore Police Force NPCC Badge (2018).

SKILLS SUMMARY

Electronics

- Microcontrollers (sensor interfacing)
- PCB prototyping & soldering (basic; LED board)

Mechanical & Fabrication

- CAD (SolidWorks, Autodesk Inventor, Fusion 360)
- CNC machining (milling, lathe)
- 3D printing (FDM, SLA)

Programming: Python (Intermediate), C (Intermediate), G-code (Basic)

Languages: English (Native), Indonesian (Intermediate), Japanese (Basic)